PEER REVIEW HISTORY

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ARTICLE DETAILS

TITLE (PROVISIONAL)	Examining family planning and adverse pregnancy outcomes for women with active tuberculosis disease: A systematic review
AUTHORS	Nguyen, Yen; McNabb, Katherine C.; Farley, Jason E.; Warren, Nicole

VERSION 1 – REVIEW

REVIEWER	Odayar, Jasantha University of Cape Town
REVIEW RETURNED	25-Jul-2021

GENERAL COMMENTS	Reviewer's report ---------- - Major Revisions Background: You say in the abstract and background that TB has long been associated with poor pregnancy outcomes. Its thus not clear why a review of pregnancy outcomes is required. Background: Are there any drug interactions between TB treatment and contraceptives and if so how would this complicate issues? Eligibility criteria: Please justify/explain why articles before 2009 were excluded. It was difficult to work out whether the review was addressing drug susceptible or drug resistant TB. Would help if it was made clearer in the aims that this will look at dru-susceptible and drug resistant TB. Methods, Eligibility criteria: Please define/specify the outcomes you were looking at i.e. pregnancy/maternal or fetal and birth outcomes (i.e. LBW, preterm etc). Also prior to this you have been saying pregnancy outcomes only. The same goes for TB treatment outcomes. Methods, Search and Identification Process: It sounds here as if one search was done for pregnancy outcomes and family planning. Looking at search terms, this is not so – please clarify. Results: Are there any studies in which pregnant women were diagnosed with TB vs studies in which women with TB fell pregnant? Results, Fetal/newborn outcomes: “LBW (weight of less than 2,500 grams at birth) was a common fetal outcome for TB infected women with or without HIV coinfection across multiple studies.” What type of studies? How many participants? Can you give percentages?

	"Preterm labor, which adds to the risk of LBW, was reported in two studies for women with TB infection." Same as above – type of studies, were there comparison groups, number of participants, what percentages etc had preterm labour or are there measures of association you can report? How was preterm defined? Results, Maternal outcomes: Please can you give an idea of effect sizes e.g. anaemia and postpartum haemorrhage more common – can you give us an idea of estimates or a range of estimates? Same for the other outcomes e.g. maternal and foetal mortality with TB HIV co-infection. Results: Under eligibility criteria you mentioned TB outcomes but these are not specified in the results? Results, HIV coinfection outcome and ART treatment: " In studies that included participants with HIV coinfection, increased maternal mortality and fetal mortality were reported." Compared to? Studies that didn't include women with HIV or were their individuals in these studies without HIV i.e. a comparison group? Results, HIV coinfection outcome and ART treatment: Regarding ART in pregnant women with TB – where and when were these studies? Are there studies comparing outcomes when on ART vs not on ART? Results: Please provide more detail on how outcomes were defined especially since the heterogeneity made it impossible to do a meta-analysis. Need more info on the heterogeneity. Results: Can you provide more detail on the process of developing the guidelines, especially considering the lack of data? Results, HIV coinfection outcome and ART treatment: "In studies that included participants with HIV coinfection, increased maternal mortality and fetal mortality were reported." Same as above – types of studies, comparison groups, effect sizes etc. "Other pregnancy outcomes included anemia, preeclampsia, LBW, and fetal mortality." What about these? More common, less common etc? Discussion: Since there are so few data on FP in women on TB treatment, what research do you think needs to be done to inform future guidelines? I see that this is in the recommendations but would be good to mention in the discussion and to be a bit more specific. Not sure why 13c and 18 of the PRISMA checklist are NA? Why was quality assessment/risk of bias not included? - Minor Revisions

REVIEWER	Adjobimey, Menonli Centre National Hospitalier Universitaire de Pneumo-Pthisiologie de Cotonou, URESTE
REVIEW RETURNED	26-Sep-2021

GENERAL COMMENTS	The research does not respect the picot approach (population, intervention, comparator, outcome, time) necessary for systematics reviews 3- The qualitative study of the articles is not well specified

	4- The articles included are of very low quality in terms of classification for systematics reviews (case report, case series, case-control, secondary data analysis). series, case-control, secondary data analysis) 5- The second part of the recommendation2 is not clear and does not present itself as a recommendation 6- Discussion not consistent 7- About the Prisma flow diagram it will be better to separate items Identification, Screening, Eligibility, and Inclusion 8- Why are there duplicate items in the eligibility section when duplicates should be before screening 9- Biases related to original articles have not been discussed 10- The composition of the research team has not been mentioned 11- Abstract : place of objective why not after introduction

VERSION 1 – AUTHOR RESPONSE

Reviewer 1			
	be clear about objectives	We have clarified our objectives in the abstract and background section of the manuscript.	Page 2 Page 6
	define outcomes more clearly	Outcomes have been clarified under the inclusion/exclusion criteria section.	Page 5-6
	have more detail in the results (when you say outcomes are worse, how much worse??)	We have updated the main manuscript to add more details to our results as well. Additional details including effect sizes and statistical results are included in Supplemental Table 2.	Page 8-14 Supplemental Table 2
	describe more of the heterogeneity in outcomes e.g. so many studies defined preterm in this way and so many defined it in that way.	Thank you for your suggestion. We have further described the heterogeneity of study designs and outcomes in our limitations section.	Page 17
Reviewer 2			
	The research does not respect the picot approach (population, intervention, comparator, outcome, time) necessary for systematics reviews	Thank you so much for your comment. Rather than utilizing the PICOT approach, we used the PRISMA guidelines to conduct our systematic review.	Page 5
	3- The qualitative study of the articles is not well specified	Thank you for your attention. There were no qualitative studies identified in our search and we have clarified this in our results section. For clarity, we have categorized the studies by their level of evidence in Supplemental Table 1.	Page 8 Supplemental Table 1
	4- The articles included are of very low quality in terms of classification for systematics reviews (case report, case series, case-control, secondary data	Thank you for your feedback. We recognized this variety of evidence as a limitation of this systematic review and discussed it in our limitations section. We aimed to provide robust evidence in the clinical and research settings on	Page 17

	analysis). series, case-control, secondary data analysis)	pregnancy outcomes and family planning strategies by including all studies.	
	5- The second part of the recommendation 2 is not clear and does not present itself as a recommendation	Thank you for your feedback. We have reviewed and updated the recommendation in Table 1. Recommendations for Integrated TB/DR-TB and Family Planning Services to ensure clarity of our recommendation.	Page 17 Table 1
	6- Discussion not consistent	Thank you for your feedback. We have re-categorized the discussion to reflect more consistently the results of this systematic review.	Page 13
	7- About the Prisma flow diagram it will be better to separate items Identification, Screening, Eligibility, and Inclusion	We followed the PRISMA flow diagram format per the PRISMA guidelines and as you recommended.	Figure 1. PRISMA flow diagram
	8- Why are there duplicate items in the eligibility section when duplicates should be before screening	We followed the PRISMA guidelines when identifying duplicates. As indicated in Figure 1, duplicates were removed before screening. During the eligibility assessment, we identified further 1 duplicate of article not captured by Covidence in the identification process. We aimed to be as transparent in our search as possible and thus provided reason during the full-text screening for exclusion of 1 duplicate article.	Figure 1. PRISMA flow diagram
	9- Biases related to original articles have not been discussed	Quality and biases related to the included articles are further detailed in our limitations section. We have also included Supplemental Table 1 detailing the quality of each study.	Page 17, Supplemental Table 1
	10- The composition of the research team has not been mentioned	The composition and contribution of the members of our research team were discussed under the acknowledgments section.	Page 18
	11- Abstract: place of objective why not after introduction	We have fixed the abstract to reflect the PRISMA guidelines for abstract. The introduction has been removed.	Page 2 (abstract)
Reviewer 3			
	Background: You say in the abstract and background that TB has long been associated with poor pregnancy outcomes. Its thus not clear why a review of pregnancy outcomes is required.	Thank you for your attention. We have removed this sentence as it is more appropriately placed in the discussion section.	Page 5
	Background: Are there any drug interactions between TB treatment	We have expanded on this thought in the background section.	Page 5

	and contraceptives and if so how would this complicate issues?		
	Eligibility criteria: Please justify/explain why articles before 2009 were excluded.	We excluded articles prior to 2009 to include only the most up-to-date data on the topic.	Page 6
	It was difficult to work out whether the review was addressing drug susceptible or drug resistant TB. Would help if it was made clearer in the aims that this will look at drug susceptible and drug resistant TB.	Thank you so much for your feedback. We have clarified that this study looked at both drug susceptible and drug resistant TB.	Page 4-5
	Methods, Eligibility criteria: Please define/specify the outcomes you were looking at i.e. pregnancy/maternal or fetal and birth outcomes (i.e. LBW, preterm etc). Also prior to this you have been saying pregnancy outcomes only. The same goes for TB treatment outcomes.	Thank you so much for your response. Outcomes have been clarified under the inclusion/exclusion criteria section.	Page 5-6
	Methods, Search and Identification Process: It sounds here as if one search was done for pregnancy outcomes and family planning. Looking at search terms, this is not so – please clarify.	We have updated our search and clarified in the appendix as well as the manuscript. Only one search was done looking at pregnancy outcomes and family planning.	Page 6 Appendix
	Results: Are there any studies in which pregnant women were diagnosed with TB vs studies in which women with TB fell pregnant?	There were a few studies which identified both women who became pregnant during treatment and those who were pregnant prior to TB treatment; the details are in Supplemental Table 2.	Supplemental Table 2
	Results, Fetal/newborn outcomes: o “LBW (weight of less than 2,500 grams at birth) was a common fetal outcome for TB infected women with or without HIV coinfection across multiple studies.” What type of studies? How many participants? Can you give percentages? o “Preterm labor, which adds to the risk of LBW, was reported in two studies for women with TB infection.” Same as above – type of studies, were there comparison groups, number of participants, what percentages etc had preterm labour or are there measures of	We have expanded on our results to address your feedback. The details you requested are also available to peruse on the Supplemental Table 2.	Page 8-9

	association you can report? How was preterm defined?		
	Results, Maternal outcomes: Please can you give an idea of effect sizes e.g. anaemia and postpartum haemorrhage more common – can you give us an idea of estimates or a range of estimates? Same for the other outcomes e.g. maternal and foetal mortality with TB HIV co-infection.	Thank you so much for your feedback. We have provided the requested information to expand on our findings.	Page 9-11 Supplemental Table 2
	Results: Under eligibility criteria you mentioned TB outcomes but these are not specified in the results?	Thank you for your attention. TB outcomes are specified under fetal/newborn outcomes and maternal outcomes.	Page 9 Page 10
	Results, HIV coinfection outcome and ART treatment: "In studies that included participants with HIV coinfection, increased maternal mortality and fetal mortality were reported." Compared to? Studies that didn't include women with HIV or were their individuals in these studies without HIV i.e. a comparison group?	We have added statistical results and clarified the studies' samples per your recommendation.	Page 11-12
	Results, HIV coinfection outcome and ART treatment: Regarding ART in pregnant women with TB – where and when were these studies? Are there studies comparing outcomes when on ART vs not on ART?	Thank you for your feedback. We have responded to your inquiries in the manuscript.	Page 12
	Results: Please provide more detail on how outcomes were defined especially since the heterogeneity made it impossible to do a meta-analysis. Need more info on the heterogeneity.	Thank you for your suggestion. We have further described the heterogeneity of study designs and outcomes in our limitations section.	Page 17
	Results: Can you provide more detail on the process of developing the guidelines, especially considering the lack of data? o Results, HIV coinfection outcome and ART treatment: "In studies that included participants with HIV coinfection, increased maternal mortality and fetal mortality were reported." Same as above – types of studies, comparison groups, effect sizes etc. o "Other pregnancy outcomes included anemia, preeclampsia,	Thank you for your feedback. The details of each individual study (effect size, type of study, comparison group, etc.) are discussed in the Supplemental Table 2. We have also added new details in the results section per your recommendations to provide clarity. Based on our search of the literature, there are consistent evidence supporting our recommendations to integrate TB/DR-TB and family planning services. We also incorporated best practice from the field of family planning, specifically from WHO family planning	Page 14 Supplemental Table 2

	LBW, and fetal mortality." What about these? More common, less common etc?	recommendations and the United States Agency for International Development and Family Health International 360. We discussed this further in our discussion section.	
	Discussion: Since there are so few data on FP in women on TB treatment, what research do you think needs to be done to inform future guidelines? I see that this is in the recommendations but would be good to mention in the discussion and to be a bit more specific.	Thank you so much for your feedback. We have added recommendations for future research in our discussion and limitations section.	Page 14-17
	Not sure why 13c and 18 of the PRISMA checklist are NA? Why was quality assessment/risk of bias not included?	We have included a quality assessment/risk of bias in the Supplemental Table 1 for your perusal. We have also discussed potential biases of included studies in the limitations section. We have made the changes in the PRISMA checklist accordingly.	Supplemental Table 1. Critical appraisal of the included studies Page 17 PRISMA checklist